ProyectoAED

Importación de los DataFrames

```
##
## Adjuntando el paquete: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union

##
## Adjuntando el paquete: 'magrittr'

## The following object is masked from 'package:tidyr':
##
##     extract

##
## Adjuntando el paquete: 'lubridate'

## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union
```

```
poblacion <- read_tsv('data/poblacion_paises.tsv')
```

```
## Rows: 51 Columns: 27
## -- Column specification --------------------------------------------------------
## Delimiter: "\t"
## chr (27): freq,unit,age,sex,geo\TIME_PERIOD, 2000, 2001, 2002, 2003, 2004, 2...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
#str(poblacion)
#head(poblacion)

colnames(poblacion)[1] <- 'country'
poblacion %<>% mutate(country = sub("^.*,", "", country)) %>% mutate(across(-1, ~
↪  as.numeric(gsub('[^0-9.]', '', .x))))
```

```r
poblacion %<>% transmute(country, population = rowMeans(select(., -country), na.rm =
↪  TRUE))
# Calculamos la media poblacional de cada pais desde el 2000 hasta hoy

poblacion %<>% arrange(desc(population)) %>% filter(!(country %in% c('RU', 'UA',
↪  'DE_TOT', 'TR', 'FX', 'BY', 'AZ', 'UK')))
# Ordenamos los paises de mayor a menor poblacion
# Eliminamos algunos que no pertenecen politicamente a Europa

mas.poblacion10 <- poblacion$country[1:10]
# Nos quedamos con los nombres de los 10 primeros

mas.poblacion5 <- poblacion$country[1:5]
# Nos quedamos con los nombres de los 5 primeros
```

1. Autonomía residencial

```r
indep.resid <- read_tsv('data/edad_independencia_residencial.tsv')
```

```
## Rows: 99 Columns: 27
## -- Column specification --------------------------------------------------------
## Delimiter: "\t"
## chr (25): freq,unit,sex,geo\TIME_PERIOD, 2000, 2001, 2002, 2003, 2004, 2005,...
## dbl  (2): 2018, 2019
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
#str(indep.resid)
#head(indep.resid)

colnames(indep.resid)[1] <- 'sex.country'
indep.resid %<>% mutate(sex.country = sub("^.*,([^,]*,[^,]*)$", "\\1", sex.country)) %>%
↪  separate(sex.country, into = c('sex', 'country'), sep = ',') %>% mutate(sex =
↪  factor(sex))

indep.resid %<>% mutate(across(c(-1, -2), ~ as.numeric(gsub('[^0-9.]', '', .x)))) %>%
↪  pivot_longer(cols = c(-1, -2), names_to = 'year', values_to = 'age') %>% mutate(year
↪  = as.numeric(year))
```

```r
vivir.padres1834 <- read_tsv('data/vivir_con_padres18-34.tsv')
```

```
## Rows: 230 Columns: 24
## -- Column specification --------------------------------------------------------
## Delimiter: "\t"
## chr (24): freq,wstatus,age,unit,geo\TIME_PERIOD, 2003, 2004, 2005, 2006, 200...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
#str(vivir.padres1834)
#head(vivir.padres1834)

colnames(vivir.padres1834)[1] <- 'status.country'
vivir.padres1834 %<>% mutate(status.country = sub("^[^,]*,([^,]*,).*,([^,]*)$", "\\1\\2",
↪  status.country)) %>% separate(status.country, into = c('status', 'country'), sep =
↪  ',') %>% mutate(status = factor(status, levels = c('EMP_FT', 'EMP_PT', 'UNE', 'EDUC',
↪  'INAC_OTH'), labels = c('full-time', 'part-time', 'unemployed', 'student', 'other')))

vivir.padres1834 %<>% mutate(across(c(-1, -2), ~ as.numeric(gsub('[^0-9.]', '', .x))))
↪  %>% pivot_longer(cols = c(-1, -2), names_to = 'year', values_to = 'rate') %>%
↪  mutate(year = as.numeric(year))
```

```r
precio.vivienda <- read_tsv('data/precio_vivienda.tsv')
```

```
## Rows: 37 Columns: 87
## -- Column specification --------------------------------------------------------
## Delimiter: "\t"
## chr (87): freq,purchase,unit,geo\TIME_PERIOD, 2005-Q1, 2005-Q2, 2005-Q3, 200...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
#str(precio.vivienda)
#head(precio.vivienda)

colnames(precio.vivienda)[1] <- 'country'
precio.vivienda %<>% mutate(country = sub(".*,", "", country))

precio.vivienda %<>% mutate(across(-1, ~ as.numeric(gsub('[^0-9.]', '', .x)))) %>%
↪  pivot_longer(cols = -1, names_to = 'year.quadri', values_to = 'rel.price2015')

precio.vivienda %<>% separate(year.quadri, into = c('year', 'quadri'), sep = '-') %>%
↪  mutate(quadri = factor(quadri, ordered=TRUE)) %>% mutate(year = as.numeric(year))
```

```r
precio.alquiler <- read_tsv('data/precio_alquiler.tsv')
```

```
## Rows: 37 Columns: 27
## -- Column specification --------------------------------------------------------
## Delimiter: "\t"
## chr (27): freq,unit,coicop,geo\TIME_PERIOD, 2000, 2001, 2002, 2003, 2004, 20...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
#str(precio.alquiler)
#head(precio.alquiler)

colnames(precio.alquiler)[1] <- 'country'
precio.alquiler %<>% mutate(country = sub(".*,", "", country))

precio.alquiler %<>% mutate(across(-1, ~ as.numeric(gsub('[^0-9.]', '', .x)))) %>%
↪  pivot_longer(cols = -1, names_to = 'year', values_to = 'rel.price2015') %>%
↪  mutate(year = as.numeric(year))
```

```r
salarios <- read_tsv('data/salarios.tsv')
```

```
## Rows: 93 Columns: 26
## -- Column specification --------------------------------------------------------
## Delimiter: "\t"
## chr (26): freq,age,sex,statinfo,unit,geo\TIME_PERIOD, 2000, 2001, 2003, 2004...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
#str(salarios)
#head(salarios)

colnames(salarios)[1] <- 'age.country'
salarios %<>% mutate(age.country = sub("^[^,]*,([^,]*,).*,([^,]*)$", "\\1\\2",
↪  age.country)) %>% separate(age.country, into = c('age', 'country'), sep = ',') %>%
↪  mutate(age = factor(age, ordered = TRUE))

salarios %<>% mutate(across(c(-1, -2), ~ as.numeric(gsub('[^0-9.]', '', .x)))) %>%
↪  pivot_longer(cols = c(-1, -2), names_to = 'year', values_to = 'price.eur') %>%
↪  mutate(year = as.numeric(year))
```

```r
ipca <- read_tsv('data/ipca.tsv')
```

```
## Rows: 46 Columns: 27
## -- Column specification --------------------------------------------------------
## Delimiter: "\t"
## chr (27): freq,unit,coicop18,geo\TIME_PERIOD, 2000, 2001, 2002, 2003, 2004, ...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
#str(ipca)
#head(ipca)

colnames(ipca)[1] <- 'country'
ipca %<>% mutate(country = sub(".*,", "", country))

ipca %<>% mutate(across(-1, ~ as.numeric(gsub('[^0-9.]', '', .x)))) %>% pivot_longer(cols
↪  = -1, names_to = 'year', values_to = 'rel.value2025') %>% mutate(year =
↪  as.numeric(year))
```

```r
hacinamiento1629 <- read_tsv('data/hacinamiento_nacimiento16-29.tsv')
```

```
## Rows: 408 Columns: 24
## -- Column specification --------------------------------------------------------
## Delimiter: "\t"
## chr (24): freq,unit,c_birth,sex,age,geo\TIME_PERIOD, 2003, 2004, 2005, 2006,...
```

```
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
#str(hacinamiento1629)
#head(hacinamiento1629)

colnames(hacinamiento1629)[1] <- 'birth.sex.country'
hacinamiento1629 %<>% mutate(birth.sex.country = sub("^.*,([^,]*,[^,]*,)[^,]*,([^,]*)$",
↪  "\\1\\2", birth.sex.country)) %>% separate(birth.sex.country, into = c('birth',
↪  'sex', 'country'), sep = ',') %>% mutate(birth = factor(birth, levels = c('EU28_FOR',
↪  'NEU28_FOR', 'NAT'), labels = c('European', 'Foreigner', 'National'))) %>% mutate(sex
↪  = factor(sex))

hacinamiento1629 %<>% mutate(across(c(-1, -2, -3), ~ as.numeric(gsub('[^0-9.]', '',
↪  .x)))) %>% pivot_longer(cols = c(-1, -2, -3), names_to = 'year', values_to = 'rate')
↪  %>% mutate(year = as.numeric(year))
```

```
retr.pagos1629 <- read_tsv('data/retrasos_pagos_16-29.tsv')
```

```
## Rows: 46 Columns: 17
## -- Column specification --------------------------------------------------------
## Delimiter: "\t"
## chr (17): freq,unit,lev_limit,age,sex,geo\TIME_PERIOD, 2010, 2011, 2012, 201...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
#str(retr.pagos1629)
#head(retr.pagos1629)

colnames(retr.pagos1629)[1] <- 'country'
retr.pagos1629 %<>% mutate(country = sub("^.*,", "", country))

retr.pagos1629 %<>% mutate(across(-1, ~ as.numeric(gsub('[^0-9.]', '', .x)))) %>%
↪  pivot_longer(cols = -1, names_to = 'year', values_to = 'rate') %>% mutate(year =
↪  as.numeric(year))
```

2. Seguridad económica

```
tasa.desemp1539 <- read_tsv('data/desempleo_nivel_educativo15-39.tsv')
```

```
## Rows: 342 Columns: 107
## -- Column specification --------------------------------------------------------
## Delimiter: "\t"
## chr (107): freq,unit,sex,age,isced11,geo\TIME_PERIOD, 2000-Q1, 2000-Q2, 2000...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# Cargamos los datos
#str(tasa.desemp1539)
#head(tasa.desemp1539)
# Los visualizamos para ver problemas

colnames(tasa.desemp1539)[1] <- 'sex.level.country'
# Renombramos la 1a columna, ya que solo nos interesan el sexo, nivel educativo y pais
tasa.desemp1539 %<>% mutate(sex.level.country = sub("^.*,([^,]*),([^,]*),([^,]*,[^,]*)$",
↪  "\\1,\\3", sex.level.country)) %>% separate(sex.level.country, into = c('sex',
↪  'level', 'country'), sep = ',') %>% mutate(level = factor(level, levels = c('ED0-2',
↪  'ED3_4', 'ED5-8'), labels = c('until.mid.sec', 'until.bac', 'university'), ordered =
↪  TRUE)) %>% mutate(sex = factor(sex))
# Quitamos la informacion de las instancias salvo sexo, nivel educativo y pais
# Y separamos ambas columnas
# Convertimos el sexo y el nivel educativo en factores

tasa.desemp1539 %<>% mutate(across(c(-1, -2, -3), ~ as.numeric(gsub('[^0-9.]', '', .x))))
↪  %>% pivot_longer(cols = c(-1, -2, -3), names_to = 'year.quadri', values_to = 'rate')
# Hacemos que las tasas sean exclusivamente numericas
# Pasamos los años a formato tidy
# Obtenemos una columna con los años y sus respectivos cuatrimestres

tasa.desemp1539 %<>% separate(year.quadri, into = c('year', 'quadri'), sep = '-') %>%
↪  mutate(quadri = factor(quadri, ordered=TRUE)) %>% mutate(year = as.numeric(year))
# Separamos en columnas de año y cuatrimestre
# Pasamos los cuatrimestres a factor
# Pasamos los años a numerico
```

```r
tempor.inv1529 <- read_tsv('data/temporalidad_involuntaria15-29.tsv')
```

```
## Rows: 444 Columns: 27
## -- Column specification --------------------------------------------------------
## Delimiter: "\t"
## chr (27): freq,unit,sex,age,reason,geo\TIME_PERIOD, 2000, 2001, 2002, 2003, ...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
#str(tempor.inv1529)
#head(tempor.inv1529)

colnames(tempor.inv1529)[1] <- 'sex.country'
tempor.inv1529 %<>% mutate(sex.country = sub("^.*,([^,]*),([^,]*,[^,]*),([^,]*)$",
↪  "\\1,\\3", sex.country)) %>% separate(sex.country, into = c('sex', 'country'), sep =
↪  ',') %>% mutate(sex = factor(sex))

tempor.inv1529 %<>% mutate(across(c(-1, -2), ~ as.numeric(gsub('[^0-9.]', '', .x)))) %>%
↪  pivot_longer(cols = c(-1, -2), names_to = 'year', values_to = 'rate') %>% mutate(year
↪  = as.numeric(year))
```

```r
tempor.educ1539 <- read_tsv('data/temporalidad_nivel_educativo15-39.tsv')
```

```
## Rows: 342 Columns: 27
## -- Column specification --------------------------------------------------------
## Delimiter: "\t"
## chr (27): freq,unit,sex,age,isced11,geo\TIME_PERIOD, 2000, 2001, 2002, 2003,...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

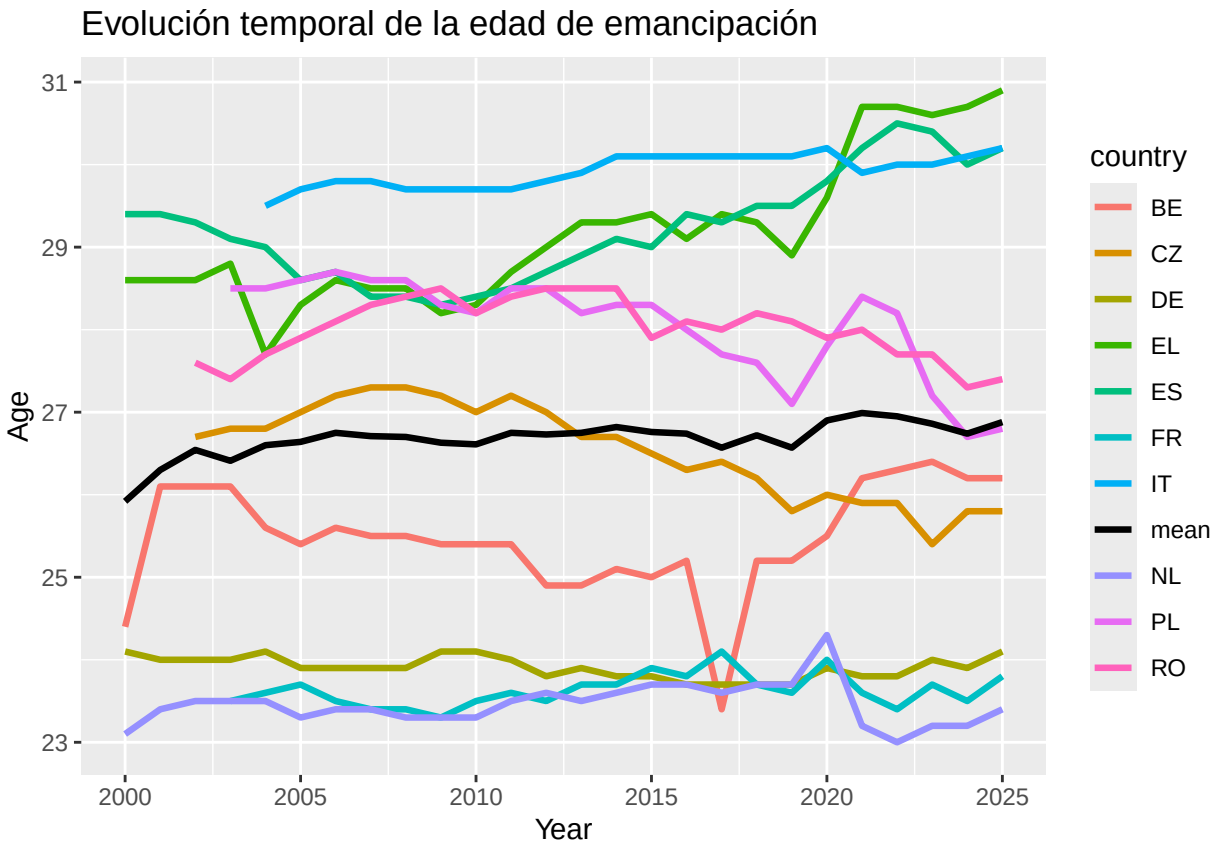
```r
#str(tempor.educ1539)
#head(tempor.educ1539)

colnames(tempor.educ1539)[1] <- 'sex.level.country'
tempor.educ1539 %<>% mutate(sex.level.country = sub("^.*,([^,]*),([^,]*),([^,]*,[^,]*)$",
↪  "\\1,\\3", sex.level.country)) %>% separate(sex.level.country, into = c('sex',
↪  'level', 'country'), sep = ',') %>% mutate(level = factor(level, levels = c('ED0-2',
↪  'ED3_4', 'ED5-8'), labels = c('until.mid.sec', 'until.bac', 'university'), ordered =
↪  TRUE)) %>% mutate(sex = factor(sex))

tempor.educ1539 %<>% mutate(across(c(-1, -2, -3), ~ as.numeric(gsub('[^0-9.]', '', .x))))
↪  %>% pivot_longer(cols = c(-1, -2, -3), names_to = 'year', values_to = 'thous.pers')
↪  %>% mutate(year = as.numeric(year))
```

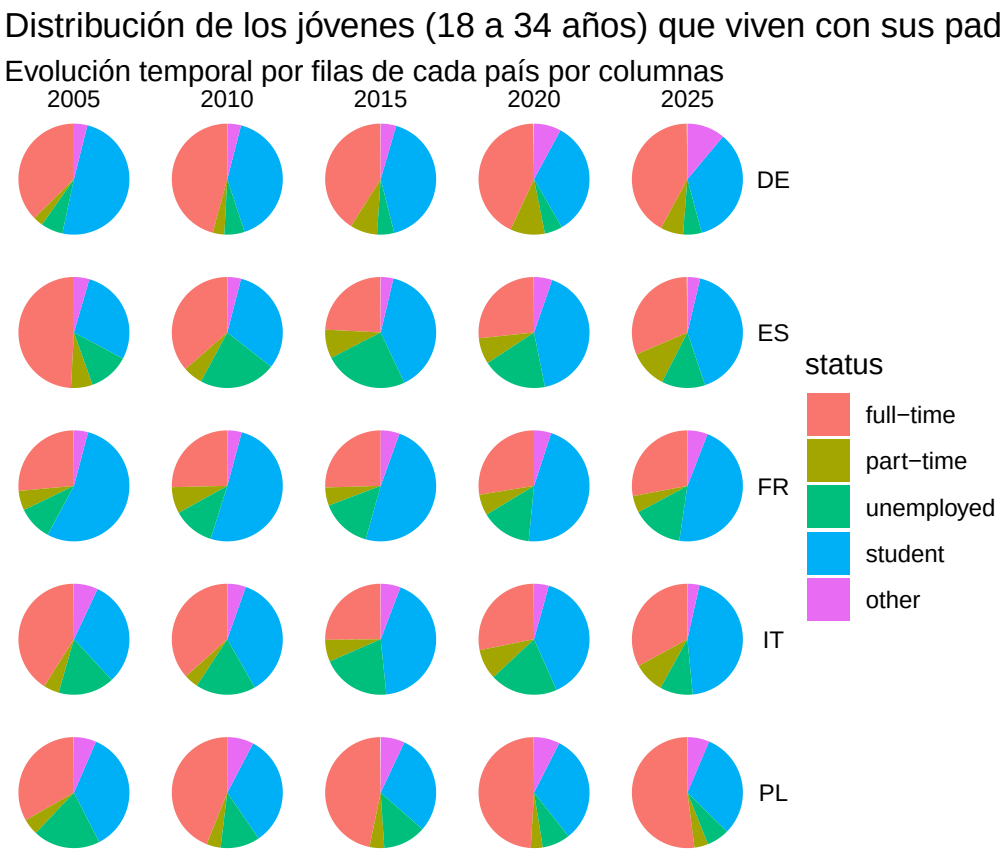
Análisis visual

```r
indep.resid.total <- indep.resid %>% filter(sex == 'T' & country %in% mas.poblacion10)
age.mean <- indep.resid.total %>% group_by(year) %>% summarise(age.mean = mean(age, na.rm
↪  = TRUE), .groups = "drop")
ggplot() + geom_line(data = indep.resid.total, mapping = aes(x = year, y = age, color =
↪  country), linewidth = 1.15, na.rm = TRUE) + geom_line(data = age.mean, mapping =
↪  aes(x = year, y = age.mean, color = 'mean'), linewidth = 1.15, na.rm = TRUE) +
  scale_color_manual(values =
  ↪  c(setNames(scales::hue_pal()(length(unique(indep.resid.total$country))),
  ↪  unique(indep.resid.total$country)), "mean" = "black")) +
  labs(title = 'Evolución temporal de la edad de emancipación') + xlab('Year') +
  ↪  ylab('Age')
```




```
ggsave('results/edad_emancipacion.pdf', width = 14, height = 10, dpi = 300)
```

```
ggplot(filter(vivir.padres1834, year %in% c(2005, 2010, 2015, 2020, 2025), country %in%
↪  mas.poblacion5), aes(x = "", y = rate, fill = status)) + geom_col(width = 1) +
↪  coord_polar(theta = "y") + facet_grid(country ~ year) + theme_void() + labs(title =
↪  'Distribución de los jóvenes (18 a 34 años) que viven con sus padres', subtitle =
↪  'Evolución temporal por filas de cada país por columnas')
```




```
ggsave('results/vivir_con_padres.pdf', width = 14, height = 10, dpi = 300)
```

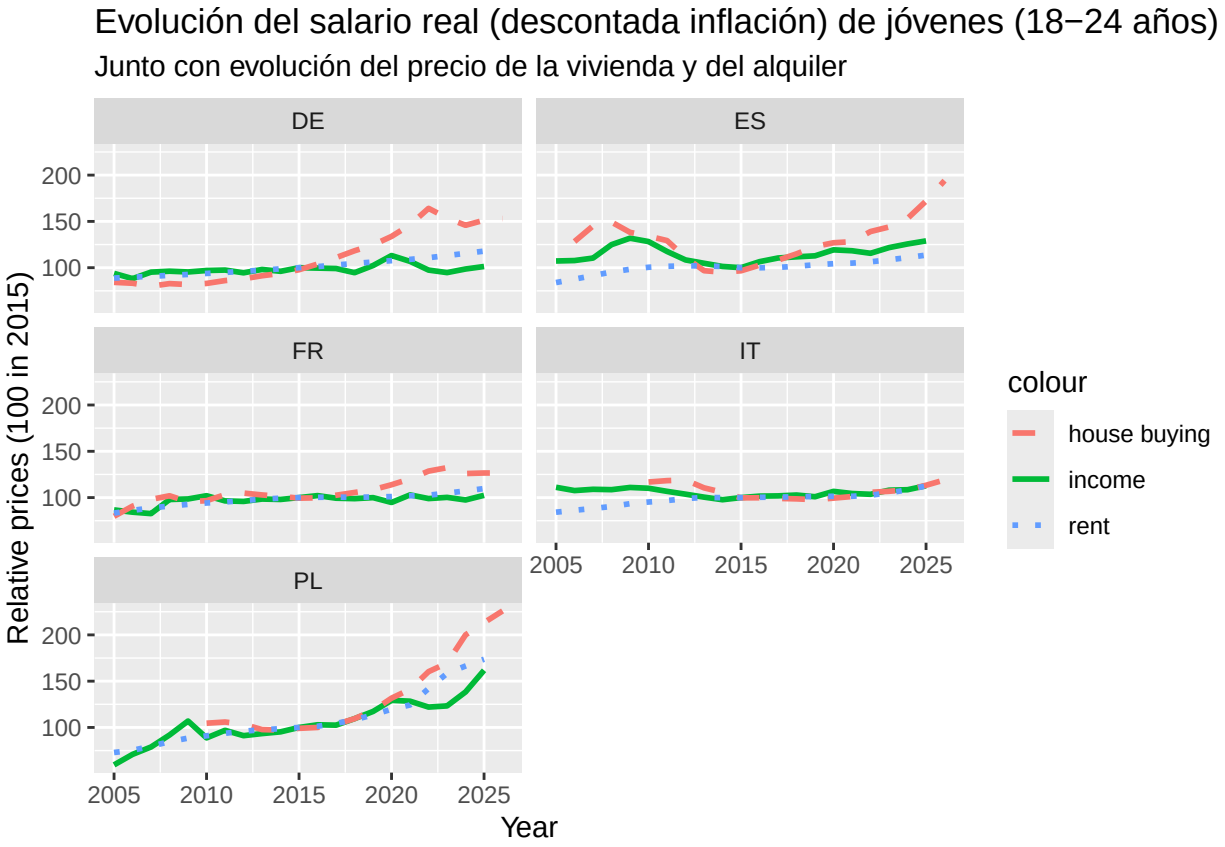
```
salarios.final <- filter(salarios, country %in% mas.poblacion5) %>%
↪  pivot_wider(names_from = age, values_from = price.eur)
ipca.final <- filter(ipca, country %in% mas.poblacion5)
salarios.infl <- right_join(ipca.final, salarios.final)
```

```
## Joining with `by = join_by(country, year)`
```

```
salarios.infl %<>% pivot_longer(cols = c(4,5), names_to = 'age', values_to = 'price.eur')
# Hacemos join de los dataframes de salarios y de la inflacion anualizada (IPCA)
salarios.infl %<>% group_by(age, country) %>% transmute(year, age, country,
↪  real.price.eur = price.eur*rel.value2025[year==2015]/rel.value2025) %>% ungroup()
# Creamos una columna con los salarios reales tomando como referencia 2015
salarios.infl %<>% group_by(age, country) %>% transmute(year, age, country, rel.price2015
↪  = real.price.eur/real.price.eur[year == 2015]*100) %>% ungroup()
# Convertimos los salarios a una variable con referencia = 100 en 2015
# Lo hacemos para poder compararlos con la evolucion de vivienda y alquiler
```

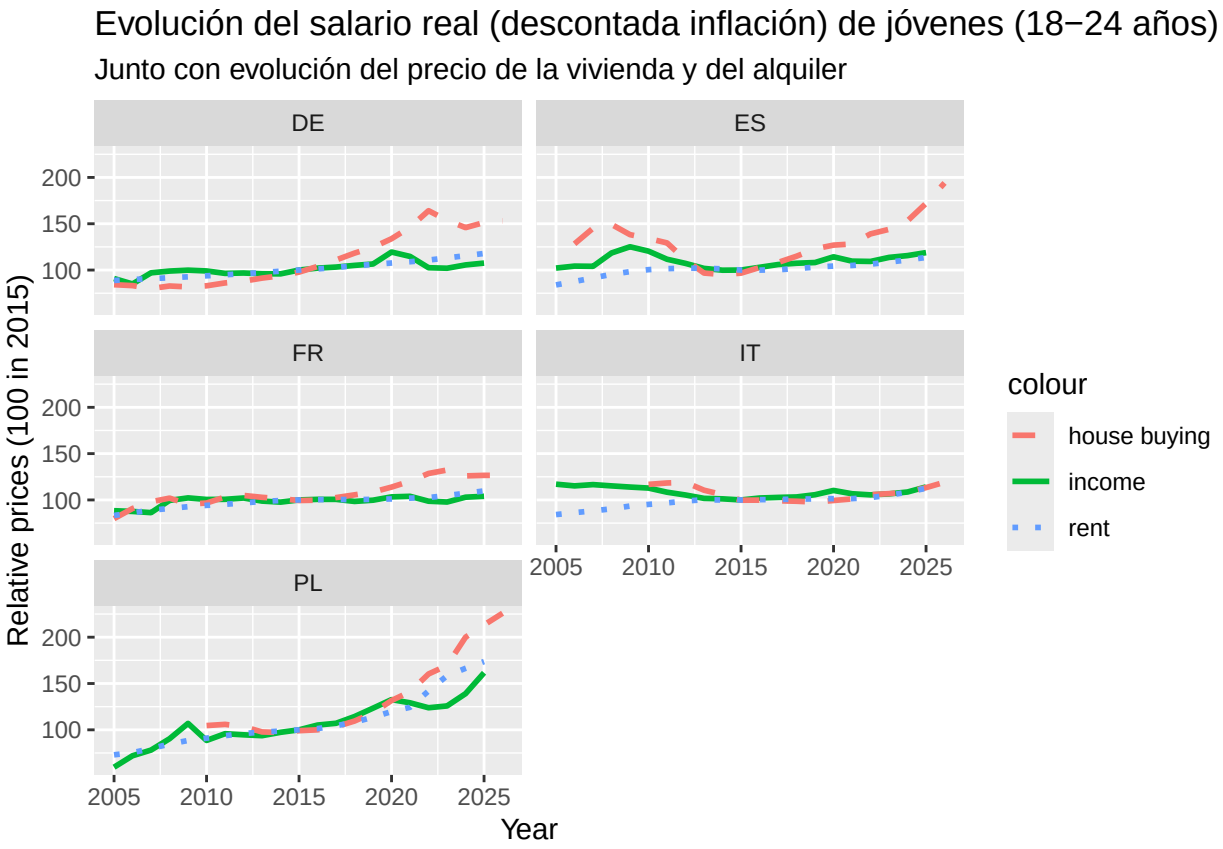
```
ggplot() +
  geom_line(data = filter(salarios.infl, age == 'Y18-24', year >= 2005), mapping = aes(x
  ↪  = year, y = rel.price2015, color = 'income'), linewidth = 1, na.rm = TRUE) +
  geom_line(data = filter(precio.vivienda, country %in% mas.poblacion5, quadri == 'Q1'),
  ↪  mapping = aes(x = year, y = rel.price2015, color = 'house buying'), linewidth = 1,
  ↪  linetype = 2, na.rm = TRUE) +
```

```
  geom_line(data = filter(precio.alquiler, country %in% mas.poblacion5, year >= 2005),
  ↪  mapping = aes(x = year, y = rel.price2015, color = 'rent'), linewidth = 1, linetype
  ↪  = 3, na.rm = TRUE) + facet_wrap(.~country, nrow = 3) + labs(title = 'Evolución del
  ↪  salario real (descontada inflación) de jóvenes (18-24 años)', subtitle = 'Junto con
  ↪  evolución del precio de la vivienda y del alquiler') + xlab('Year') +
  ↪  ylab('Relative prices (100 in 2015)')
```




```
ggsave('results/viviendas_salarios18-24.pdf', width = 14, height = 10, dpi = 300)
```

```
ggplot() +
  geom_line(data = filter(salarios.infl, age == 'Y25-49', year >= 2005), mapping = aes(x
  ↪  = year, y = rel.price2015, color = 'income'), linewidth = 1, na.rm = TRUE) +
  geom_line(data = filter(precio.vivienda, country %in% mas.poblacion5, quadri == 'Q1'),
  ↪  mapping = aes(x = year, y = rel.price2015, color = 'house buying'), linewidth = 1,
  ↪  linetype = 2, na.rm = TRUE) +
  geom_line(data = filter(precio.alquiler, country %in% mas.poblacion5, year >= 2005),
  ↪  mapping = aes(x = year, y = rel.price2015, color = 'rent'), linewidth = 1, linetype
  ↪  = 3, na.rm = TRUE) + facet_wrap(.~country, nrow = 3) + labs(title = 'Evolución del
  ↪  salario real (descontada inflación) de jóvenes (18-24 años)', subtitle = 'Junto con
  ↪  evolución del precio de la vivienda y del alquiler') + xlab('Year') +
  ↪  ylab('Relative prices (100 in 2015)')
```




```
ggsave('results/viviendas_salarios25-49.pdf', width = 14, height = 10, dpi = 300)
```

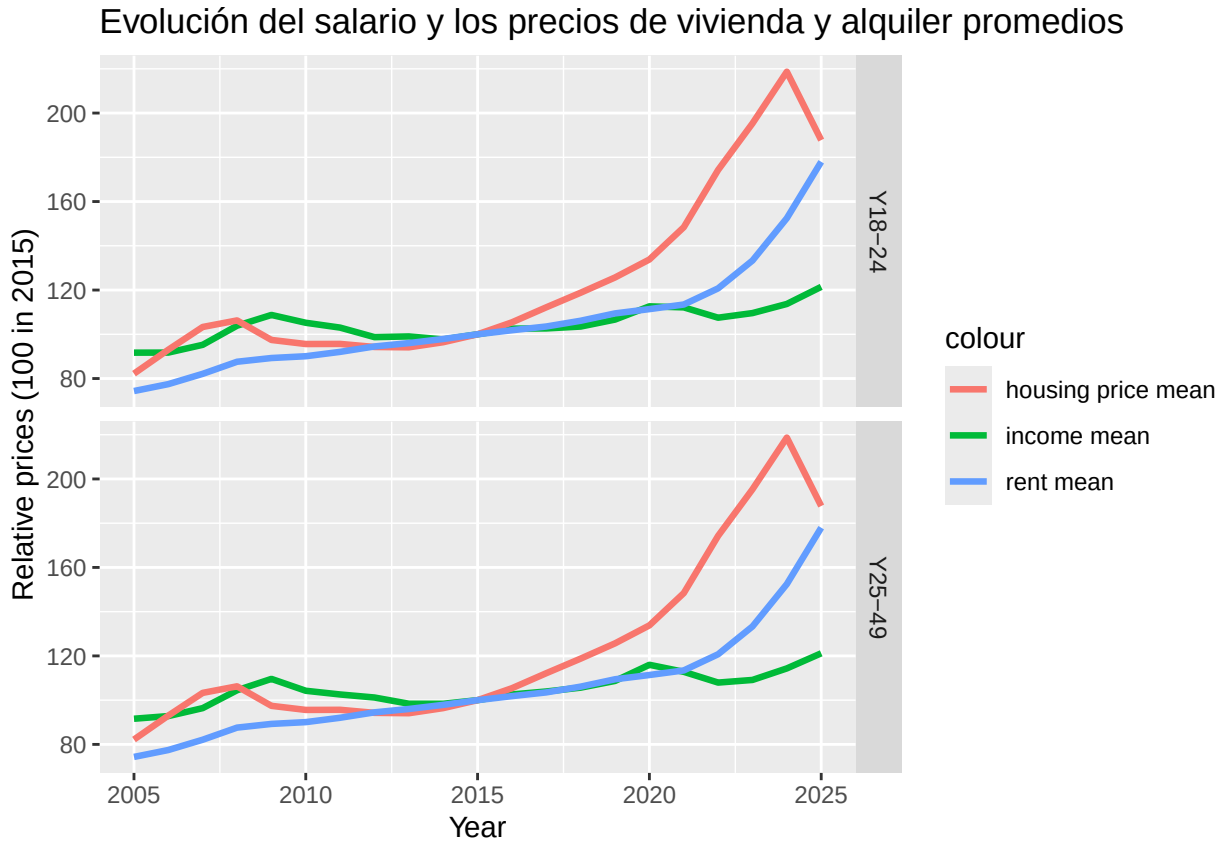
```
income.mean <- salarios.infl %>% group_by(year, age) %>% summarise(income.mean =
↪  mean(rel.price2015, na.rm = TRUE), .groups = "drop") %>% ungroup()

housing.mean <- precio.vivienda %>% group_by(year) %>% summarise(housing.mean =
↪  mean(rel.price2015, na.rm = TRUE), .groups = "drop") %>% ungroup()

rent.mean <- precio.alquiler %>% group_by(year) %>% summarise(rent.mean =
↪  mean(rel.price2015, na.rm = TRUE), .groups = "drop") %>% ungroup()

mean.df <- income.mean %>% left_join(housing.mean, by = 'year') %>% left_join(rent.mean,
↪  by = 'year') %>% filter(year >= 2005)

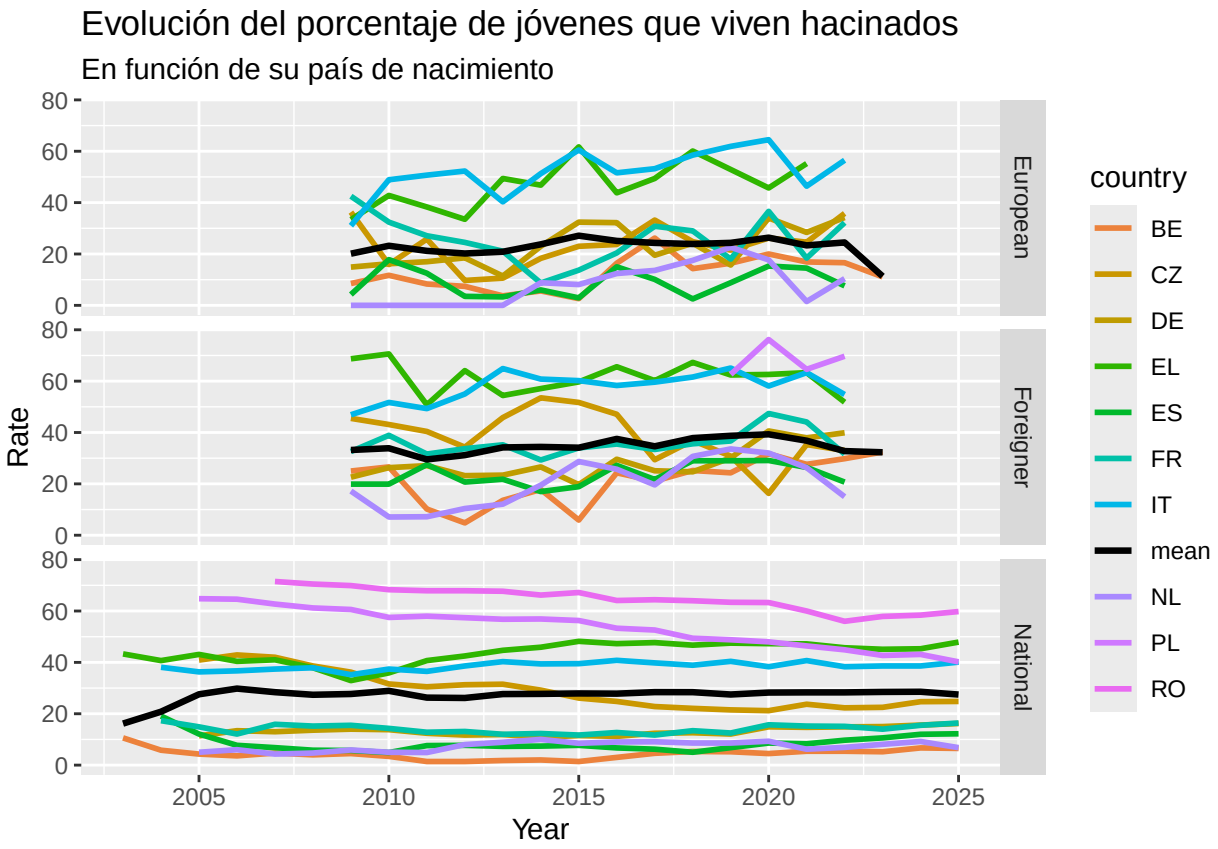
ggplot(data = mean.df) + geom_line(aes(x = year, y = income.mean, color = 'income mean'),
↪  linewidth = 1.15, na.rm = TRUE) + geom_line(aes(x = year, y = housing.mean, color =
↪  'housing price mean'), linewidth = 1.15, na.rm = TRUE) + geom_line(aes(x = year, y =
↪  rent.mean, color = 'rent mean'), linewidth = 1.15, na.rm = TRUE) + facet_grid(age~.)
↪  + labs(title = 'Evolución del salario y los precios de vivienda y alquiler
↪  promedios') + xlab('Year') + ylab('Relative prices (100 in 2015)')
```




```
ggsave('results/viviendas_salarios_medios.pdf', width = 14, height = 10, dpi = 300)
```

```
overcr.mean <- hacinamiento1629 %>% group_by(year, birth) %>% summarise(mean = mean(rate,
↪  na.rm = TRUE), .groups = "drop") %>% ungroup()

ggplot() + geom_line(data=filter(hacinamiento1629, sex == 'T', country %in%
↪  mas.poblacion10), mapping = aes(x = year, y = rate, col = country), linewidth = 1,
↪  na.rm = TRUE) + geom_line(data=overcr.mean, mapping = aes(x=year, y = mean, color =
↪  'mean'), linewidth = 1.15, na.rm = TRUE) + facet_grid(birth~.) +
  scale_color_manual(values =
   ↪  c(setNames(scales::hue_pal()(length(unique(hacinamiento1629$country))),
   ↪  unique(hacinamiento1629$country)), "mean" = "black")) +
  labs(title = 'Evolución del porcentaje de jóvenes que viven hacinados', subtitle = 'En
   ↪  función de su país de nacimiento') + xlab('Year') + ylab('Rate')
```




```
ggsave('results/hacinamiento.pdf', width = 14, height = 10, dpi = 300)
```